

Notes: Ameriks, Caplin, and Leahy (QJE, 2003)
Wealth Accumulation and the Propensity to Plan

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1 Big picture

This paper asks why households with similar observable economic and demographic characteristics accumulate very different amounts of wealth. Standard explanations based on discount factors, risk aversion, and bequest motives are theoretically natural, but previous empirical work found that such preference measures explain only a limited share of observed wealth dispersion.

The paper proposes a behavioral and skill-based explanation: households differ in their **propensity to plan**. Some households are more willing and able to formulate financial plans, monitor spending, and translate long-run saving goals into current behavior. These differences in planning attitudes and skills are strongly associated with both financial planning and wealth accumulation.

Core message. Saving may be partly a matter of skill and self-control, not only a matter of preferences. Households with a high propensity to plan accumulate more wealth because they plan more, save more, and appear better able to control spending through budgeting.

2 Incremental contribution of the paper

Main incremental contribution. The paper introduces and empirically tests the idea that a household's propensity to plan is an economically important determinant of wealth accumulation. More specifically, the paper contributes in five ways.

1. **Moves beyond standard preference explanations.** It asks whether wealth differences can be explained by attitudes and skills related to planning, rather than only by discounting, risk aversion, bequest motives, or income.
2. **Constructs new survey evidence.** The authors design two custom surveys of TIAA-CREF participants: the Survey of Participant Finances and the Survey of Financial Attitudes and Behavior. This allows them to combine detailed wealth data with behavioral questions.
3. **Uses “designer instruments.”** The paper uses nonfinancial planning traits - especially vacation planning and math confidence - as instruments for financial planning. This is meant to distinguish planning causing wealth from wealth causing planning.
4. **Tests alternative classical channels.** The analysis examines reverse causation, classical preferences, uncertainty reduction, and stock-market participation, and argues that none explains the planning-wealth relationship.
5. **Links planning to budgeting and self-control.** The paper shows that the propensity to plan is related not only to long-run financial plans but also to regular budgeting behavior, supporting an effortful self-control mechanism.

Why this is new. Earlier research, especially Lusardi's work, showed that thinking about retirement is correlated with wealth. Ameriks, Caplin, and Leahy go further by designing survey questions that measure underlying planning attitudes and skills and by using them to identify a planning channel.

3 Conceptual framework

3.1 The paper’s preferred mechanism: propensity to plan

The authors define the propensity to plan as a set of attitudes and skills that affects how a household approaches financial planning. The mechanism has two components.

1. **Control problems.** Households may want to save for the future but find it difficult to translate that long-run desire into current spending decisions. This can arise from present bias, temptation, inertia, inattention, or fixed costs of monitoring wealth.
2. **Control skills.** Households differ in their ability and willingness to monitor financial behavior, form plans, and correct overspending. These skills can include numeracy, organization, and attention to long-run goals.

In this view, high planners notice overspending earlier and adjust more easily. Low planners may let excessive spending continue for longer. The practical implication is that wealth accumulation depends partly on the household’s ability to implement its desired savings behavior.

3.2 Alternative theories considered

The paper explicitly considers three alternatives.

1. **Reverse causation.** Wealthier households may plan more simply because they have more wealth to manage or because planning is more enjoyable when retirement looks secure.
2. **Classical preference parameters.** Planning may proxy for patience, bequest motives, precautionary saving motives, or risk aversion. If so, planning would not be a separate causal factor.
3. **Planning reduces uncertainty.** Planners may understand future income or asset returns better, hold more stocks, earn higher returns, and therefore accumulate more wealth.

The paper’s empirical strategy is built to test whether these alternative explanations eliminate the planning effect.

4 Data

4.1 Surveys

The paper uses two surveys sent to TIAA-CREF participants.

- **Survey of Participant Finances (SPF), January 2000.** This survey measures household asset holdings, debt, retirement assets, nonretirement financial assets, real estate, and earnings.
- **Survey of Financial Attitudes and Behavior (FAB), January 2001.** This survey measures planning behavior, financial attitudes, expectations, self-reported skills, budgeting, and preference parameters.

The main universe consists of households in which neither the respondent nor the spouse or partner is at least 65 years old. The under-65 universe contains 1,191 households. The main regression sample contains 500 households with complete data, positive net worth, no life-annuity income from TIAA-CREF, and less than \$5 million in gross financial assets.

4.2 Sample features

The sample is not representative of the general U.S. population. Respondents are highly educated, disproportionately connected to academic or professional employment, and wealthier than the average household. This is important for external validity. However, the sample is useful for studying planning because it contains unusually detailed financial and behavioral data.

4.3 Data quality check

The authors validate self-reported wealth data by comparing reported TIAA-CREF balances with administrative accounting data. The log-log regression coefficient is essentially one, and the constant is close to zero. This supports the reliability of the self-reported data, although later robustness checks also use administrative TIAA-CREF data for households with small discrepancies between reports and records.

5 Empirical framework

The authors summarize their main empirical problem with a two-equation system:

$$w = \alpha_0 + \alpha_1 x + \alpha_2 p + \varepsilon_w,$$

$$p = \beta_0 + \beta_1 x + \beta_2 w + \beta_3 z + \varepsilon_p.$$

Here w is log wealth, p is financial planning, x is a set of standard controls, and z measures the propensity to plan.

The key coefficient is α_2 . If $\alpha_2 > 0$, then financial planning is associated with greater wealth after controlling for observable household characteristics. The core identification problem is that planning and wealth may be jointly determined.

5.1 Planning measures

The main planning variable is based on Question 1a:

I have spent a great deal of time developing a financial plan.

Responses are coded on a one-to-six agreement scale. The authors also use an output-based planning question asking whether the respondent has gathered household financial information, reviewed it in detail, and formulated a specific financial plan for the long-term future.

5.2 Instruments for planning

The main instruments are:

- **Vacation planning:** whether the respondent spends a great deal of time deciding where to go and what to do before going on vacation.
- **Math confidence:** whether the respondent is highly confident in mathematical skills.

The exclusion restriction is that these variables affect wealth through financial planning, not directly, once income, age, education, occupation, marital status, pension variables, and other controls are included. This is plausible but not guaranteed, and it is the main identifying assumption.

5.3 Controls

The standard controls include current income, past income, expected future income, age and age squared, employment status, occupation, education, defined-benefit plan participation for respondent and spouse, marital status, gender, and number of children.

6 Main results

1. **Planning is strongly correlated with wealth.** In OLS regressions, the planning variable is positive and highly significant for both net worth and gross financial assets. A one-standard-deviation increase in planning is associated with roughly a 20 percent increase in wealth.
2. **Propensity-to-plan instruments predict financial planning.** Vacation planning and math confidence are both positive and significant in the first stage.
3. **IV estimates are larger than OLS estimates.** The instrumented planning coefficient is 0.324 for net worth and 0.398 for gross financial assets, implying economically large wealth differences.
4. **Results survive data-quality checks.** Among households with small differences between self-reported and administrative TIAA-CREF balances, planning coefficients remain positive and close to the baseline values.
5. **Preference parameters do not explain the result.** Adding measures of discounting, precaution, risk aversion, and bequest motives does not reduce the estimated effect of planning.
6. **Stock holding does not explain the result.** More than 90 percent of regression-sample households own stocks, and planning does not meaningfully predict the share of financial wealth in stocks once controls are included.
7. **Planning is linked to saving and budgeting.** Higher propensity to plan predicts greater likelihood of spending less than employment income. It is also strongly related to household budgeting, and many budgeters report that budgeting helps restrain spending.

Interpretation. The evidence supports an effortful self-control channel. Planning seems to help households control spending and accumulate wealth, rather than merely proxying for standard preference parameters or portfolio choices.

7 Identification and interpretation

Strengths.

- The surveys were custom-designed for the research question and contain unusually detailed asset, debt, income, planning, budgeting, and preference information.
- The paper validates self-reported TIAA-CREF wealth using administrative data.
- The IV strategy directly addresses reverse causation from wealth to planning.
- The paper studies several mechanisms, not just the reduced-form wealth-planning correlation.

Limitations.

- The TIAA-CREF sample is highly educated and relatively wealthy, so external validity is limited.
- The exclusion restriction for vacation planning and math confidence is debatable. Math confidence may be correlated with unobserved ability, discipline, or lifetime resources.
- The data are cross-sectional, so the timing of planning, budgeting, saving, and wealth accumulation is difficult to establish perfectly.
- Planning is self-reported, so measurement may reflect self-perception or self-presentation as well as actual planning behavior.

Best interpretation. The paper provides strong evidence that planning-related attitudes and skills are associated with wealth accumulation and plausible evidence that planning has a causal effect. The causal claim depends on the validity of the instruments, but the paper's broader behavioral message is robust: household wealth accumulation is closely connected to planning, monitoring, and budgeting behavior.

8 Tables

8.1 Tables I and II: sample composition and household finances

Characteristic	Under-65 universe		Regression sample	
	(n)	(%)	(n)	(%)
Age				
Below 35	120	10.1	61	12.2
35-39	109	9.2	58	11.6
40-44	110	9.2	61	12.2
45-49	173	14.5	82	16.4
50-54	244	20.5	106	21.2
55-59	215	18.1	70	14.0
60-64	220	18.5	62	12.4
Gender				
Female	550	46.2	205	41.0
Male	641	53.8	295	59.0
Marital status				
Curr. married	777	65.2	332	66.4
Prev. married	193	16.2	61	12.2
Never married	221	18.6	107	21.4
Education				
College or below	342	28.7	128	25.6
Masters or prof.	466	39.1	206	41.2
PhD	383	32.2	166	33.2
Occupation				
Teaching faculty	397	33.3	162	32.4
Mgmt., sen. admn.	249	20.9	103	20.6
Other tech./prof.	304	25.5	150	30.0
Other	231	19.4	85	17.0
Number of children				
0	747	62.7	303	60.6
1	162	13.6	66	13.2
2	205	17.2	96	19.2
3+	77	6.5	35	7.0

Source. Authors' tabulations of 2000 SPF and 2001 FAB survey data.
The "under-65 universe" is all respondents to the FAB survey who were under age 65 and, if applicable and available, whose spouse reported an age of less than 65 (1191 respondents/households). Some respondents in this sample did not report data for all the above characteristics. The "regression sample" is all

Sample and measure	Mean (\$000)	Median (\$000)	Std. dev. (\$000)	# obs. (N)
Under-65 universe*				
Net worth	705	379	933	671
Gross financial assets	575	270	1,004	735
Ret. fin. assets	424	210	795	885
Nonret. fin. assets	188	45	427	938
Real estate assets	250	160	530	1145
Total debt	89	55	275	1048
Mortgage debt	79	46	259	1124
Personal debt	8	0	51	1089
1998 employment income	77	67	62	1133
1999 employment income	81	70	68	1144
Expected 2005 emp. income	87	75	82	1015
Regression sample**				
Net worth	700	394	810	500
Gross financial assets	555	306	661	500
Ret. fin. assets	390	216	450	500
Nonret. fin. assets	165	44	308	500
Real estate assets	230	153	319	500
Total debt	85	60	106	500
Mortgage debt	79	50	104	500
Personal debt	6	0	12	500
1998 employment income	81	68	62	500
1999 employment income	85	72	67	500
Expected 2005 emp. income	93	80	80	500

Source. Authors' tabulation of 2000 and 2001 survey data.
Gross financial assets is the sum of all retirement account balances, mutual funds (except real estate mutual funds), directly held stocks, directly held bonds, checking accounts, savings accounts, and CDs. "Net worth" is total assets minus mortgage debt, outstanding educational loans, outstanding personal loans, and credit card balances. All aggregates exclude the value of real estate mutual funds, whole life insurance policies, trusts, and educational savings accounts (Education IRAs and 529 plans). Respondents were instructed to provide values as of December 31, 1999. Note that these data include only the information reported by respondents on the surveys, and may therefore differ from data reported in Ameriks, Caplin, and Leahy (2002).

* For the under-65 universe, statistics are tabulated for all individuals who provided complete data for each individual item (in each row). The number of observations in each row varies, as item response varies.
** The "regression sample" members are the 500 individuals in the under-65 universe who (1) provided complete information regarding the demographic characteristics in Table I above, (2) provided complete information regarding their household's net worth, (3) provided complete information on their past, present, and expected future labor earnings, (4) have no life annuity income from TIAA-CREF, (5) have positive net worth, and (6) have less than \$5 million in gross financial assets.

Table I. Demographic characteristics of the under-65 universe and regression sample.

Table II. Financial and earnings data for surveyed households.

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- Table I shows that the regression sample contains 500 households and is highly educated: 41.2 percent have a master's or professional degree and 33.2 percent have a PhD.
- Table II shows that the regression sample is wealthy: median net worth is \$394,000 and median gross financial assets are \$306,000.
- These facts matter because the sample is excellent for observing detailed planning and retirement-related behavior, but it is not representative of all households.

8.2 Tables III and IV: data quality and planning measures

TABLE III
OLS REGRESSION: REPORTED TIAA-CREF ASSETS ON ACCOUNTING DATA

Sample & RHS variables	Coeff.	Std. err.	Pr > t
Under-65 universe			
ln(TCData)	0.995	0.008	0.000
Constant	-0.017	0.039	0.659
Regression sample			
ln(TCData)	0.997	0.009	0.000
Constant	-0.014	0.047	0.761

Source. Authors' calculations using 2001 survey data and 1999 accounting data.
This is a log-log regression of respondent's report of the value of his or her TIAA-CREF assets on actual accounting data for the respondent. For both the under-65 universe and the regression sample, the data include only those who reported and had more than \$10,000 in TIAA-CREF assets, with no immediate (payout) annuities, or TIAA-CREF IRAs, and whose reported age and gender matched the age and gender recorded in the TIAA-CREF database. For the universe, 738 observations are included in this regression; the R^2 is .958; root MSE is 0.247. For the regression sample, 381 observations are included, the R^2 is .968; root MSE is 0.014.

Table III. Validation of self-reported TIAA-CREF assets.

TABLE IV
RESPONSES TO BASIC PLANNING QUESTIONS AMONG REGRESSION SAMPLE MEMBERS

Q1a response	Q2a response					
	Has detailed plan		No detailed plan		Total	
	(n)	(%)	(n)	(%)	(n)	(%)
Disagree strongly	3	0.8	9	6.7	12	2.4
Disagree	28	7.7	47	35.1	75	15.0
Disagree somewhat	40	11.0	35	26.1	75	15.0
Agree somewhat	152	41.6	38	28.4	190	38.1
Agree	105	28.8	4	3.0	109	21.8
Agree strongly	37	10.1	1	0.7	38	7.6
Total	365	100.0	134	100.0	499	100.0

Source. Authors' tabulations of 2001 FAB survey data.
One individual in the regression sample did not respond to the detailed planning question (Q2a).

Table IV. Responses to basic planning questions.

Read:

- Table III shows that self-reported TIAA-CREF balances line up closely with administrative accounting data. The coefficient on accounting data is essentially one.
- Table IV shows that the input and output planning questions are related: households that report spending more time developing a plan are also much more likely to have a detailed plan.
- Together, these tables support the paper's measurement strategy: the wealth data are credible, and the planning variables capture meaningful variation in financial planning behavior.

8.3 Tables V and VI: OLS wealth regressions and first-stage planning regressions

TABLE V
BASIC OLS WEALTH REGRESSION RESULTS

Variable	Net worth			Gross financial assets		
	Coeff.	S.E.	Pr > t	Coeff.	S.E.	Pr > t
Planning variable	0.160***	0.035	0.000	0.186***	0.035	0.000
Log 1999 income	0.319**	0.160	0.046	0.371**	0.163	0.024
Zero 1999 income	1.860**	0.815	0.023	2.234***	0.833	0.008
Log past income	0.248	0.156	0.114	0.207	0.160	0.196
Zero past income	0.616	0.885	0.486	-0.209	0.905	0.817
Log future income	0.061	0.093	0.512	0.099	0.095	0.300
Zero future income	0.392	0.418	0.349	0.502	0.428	0.241
Age	0.211***	0.048	0.000	0.177***	0.049	0.000
Age ²	-0.001***	0.001	0.005	-0.001**	0.001	0.037
Empl. status						
Working	<i>Omitted</i>			<i>Omitted</i>		
Partially retired	0.174	0.198	0.382	0.300	0.203	0.140
Retired	0.327	0.265	0.217	0.330	0.271	0.224
Occupation						
Faculty	<i>Omitted</i>			<i>Omitted</i>		
Mgmt./sen. admin.	-0.111	0.120	0.357	-0.135	0.123	0.271
Tech./professional	0.079	0.112	0.481	0.067	0.115	0.558
Other	-0.112	0.136	0.409	-0.152	0.139	0.276
Education						
College or below	-0.197*	0.111	0.076	-0.337***	0.113	0.003
M.A./professional	<i>Omitted</i>			<i>Omitted</i>		
PhD	-0.158	0.099	0.112	-0.159	0.102	0.117
R. has DB plan	-0.138	0.100	0.170	-0.235**	0.103	0.023
S. has DB plan	-0.124	0.123	0.314	-0.246*	0.126	0.052
Marital status						
Curr. married	<i>Omitted</i>			<i>Omitted</i>		
Prev. married	-0.511***	0.144	0.000	-0.504***	0.147	0.001
Never married	-0.299**	0.128	0.020	-0.184	0.131	0.159
Male respondent	-0.010	0.090	0.912	0.067	0.092	0.462
Number of kids	0.044	0.047	0.351	0.018	0.048	0.706
Constant	-3.798***	1.098	0.001	-3.567***	1.123	0.002
N	500			500		
R ²	0.577			0.569		
F	29.61			28.59		
Pr > F	0.000			0.000		

Source. Authors' calculations based on 2000 and 2001 survey data.
Dependent variables are natural logarithms of the quantities listed at head of each set of columns.
Asterisks indicate the level of statistical confidence for rejection of the hypothesis that the relevant coefficient is (independently) equal to zero: *** indicates rejection at better than a 1 percent level of confidence; ** indicates rejection at better than a 5 percent level; and * indicates rejection at better than a 10 percent level.

TABLE VI
FIRST-STAGE PLANNING/WEALTH REGRESSION RESULTS

Variable	Regression sample		
	Coeff.	S.E.	Pr > t
Vacation planning	0.166***	0.043	0.000
Math confidence	0.150***	0.044	0.001
Log 1999 income	0.138	0.207	0.504
Zero 1999 income	0.839	1.052	0.425
Log past income	0.249	0.202	0.218
Zero past income	2.058*	1.140	0.072
Log future income	-0.151	0.120	0.210
Zero future income	-0.601	0.541	0.267
Age	-0.181***	0.062	0.003
Age ²	0.002***	0.001	0.005
Empl. status			
Working	<i>Omitted</i>		
Partially retired	0.506**	0.256	0.049
Retired	0.293	0.342	0.392
Occupation			
Faculty	<i>Omitted</i>		
Mgmt./sen. admin.	0.017	0.156	0.914
Tech./professional	-0.067	0.145	0.642
Other	0.087	0.176	0.620
Education			
College or below	0.073	0.143	0.613
M.A./professional	<i>Omitted</i>		
PhD	-0.050	0.129	0.699
R. has DB plan	0.050	0.130	0.698
S. has DB plan	0.150	0.160	0.349
Marital status			
Curr. married	<i>Omitted</i>		
Prev. married	-0.117	0.186	0.529
Never married	-0.173	0.165	0.296
Male respondent	0.011	0.116	0.926
Number of kids	-0.065	0.061	0.283
Constant	5.651***	1.409	0.000
N	500		
R ²	0.139		
F	3.35		
Pr > F	0.000		

Source. Authors' calculations based on 2000 and 2001 survey data.
Dependent variable in the regression above is the answer to Question 1a, degree of agreement (1 = Disagree strongly, 2 = Disagree, 3 = Disagree somewhat, 4 = Agree somewhat, 5 = Agree, 6 = Agree strongly) with the statement "I have spent a great deal of time developing a financial plan." Asterisks indicate the level of statistical confidence for rejection of the hypothesis that the relevant coefficient is (independently) equal to zero: *** indicates rejection at better than a 1 percent level of confidence; ** indicates rejection at better than a 5 percent level; and * indicates rejection at better than a 10 percent level.

Table V. Basic OLS wealth regression results.

Table VI. First-stage planning regression results.

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- Table V shows the basic correlation: the planning coefficient is 0.160 for net worth and 0.186 for gross financial assets, both highly significant.
- Table VI shows the first stage: vacation planning and math confidence strongly predict financial planning.
- This is the empirical foundation for the IV design. The instruments are not just conceptual; they are strongly related to the planning variable in the data.

8.4 Tables VII and VIII: IV wealth effects and accounting-data robustness

TABLE VII
SECOND-STAGE PLANNING/WEALTH REGRESSION RESULTS

Variable	Net worth			Gross financial assets		
	Coeff.	S.E.	Pr > t	Coeff.	S.E.	Pr > t
Planning variable (IV)	0.324**	0.154	0.035	0.398**	0.159	0.013
Log 1999 income	0.304*	0.164	0.065	0.351**	0.170	0.040
Zero 1999 income	1.724**	0.843	0.041	2.059**	0.873	0.019
Log past income	0.190	0.168	0.259	0.133	0.174	0.447
Zero past income	0.276	0.957	0.774	-0.648	0.992	0.513
Log future income	0.085	0.098	0.385	0.129	0.101	0.201
Zero future income	0.475	0.435	0.275	0.609	0.450	0.177
Age	0.241***	0.056	0.000	0.216***	0.058	0.000
Age ²	-0.002***	0.001	0.004	-0.002**	0.001	0.016
Empl. status						
Working	<i>Omitted</i>			<i>Omitted</i>		
Partially retired	0.101	0.214	0.638	0.206	0.221	0.354
Retired	0.269	0.276	0.330	0.255	0.286	0.373
Occupation						
Faculty	<i>Omitted</i>			<i>Omitted</i>		
Mgmt./sen. admin.	-0.100	0.123	0.415	-0.122	0.127	0.338
Tech./professional	0.095	0.116	0.410	0.088	0.120	0.463
Other	-0.124	0.140	0.374	-0.167	0.145	0.249
Education						
College or below	-0.210*	0.114	0.066	-0.354***	0.118	0.003
M.A./professional	<i>Omitted</i>			<i>Omitted</i>		
PhD	-0.143	0.103	0.164	-0.140	0.106	0.189
R. has DB plan	-0.153	0.104	0.140	-0.254**	0.107	0.018
S. has DB plan	-0.145	0.128	0.258	-0.272**	0.132	0.041
Marital status						
Curr. married	<i>Omitted</i>			<i>Omitted</i>		
Prev. married	-0.502***	0.147	0.001	-0.492***	0.153	0.001
Never married	-0.274**	0.133	0.039	-0.152	0.137	0.268
Male respondent	-0.019	0.092	0.836	0.056	0.095	0.560
Number of kids	0.053	0.049	0.278	0.030	0.051	0.556
Constant	-4.947***	1.534	0.001	-5.048***	1.589	0.002
N	500			500		

Source. Authors' calculations based on 2000 and 2001 survey data.
Dependent variables are natural logarithms of the quantities listed at head of each set of columns.
Asterisks indicate the level of statistical confidence for rejection of the hypothesis that the relevant coefficient is (independently) equal to zero: *** indicates rejection at better than a 1 percent level of confidence; ** indicates rejection at better than a 5 percent level; and * indicates rejection at better than a 10 percent level.

TABLE VIII
COEFFICIENT ON IV PLANNING VARIABLE FOR HOUSEHOLDS WITH SMALL DIFFERENCES BETWEEN SELF-REPORTS AND ACCOUNTING DATA

Dependent variable	Self-reported data			TIAA-CREF data		
	Coeff.	S.E.	Pr > t	Coeff.	S.E.	Pr > t
Net worth	0.368*	0.187	0.050	0.360*	0.185	0.053
Gross financial assets	0.357**	0.178	0.046	0.344**	0.173	0.048
N	361			361		

Source. Authors' calculations based on 2000 and 2001 survey data.
Coefficients in the table above are for the planning variable in the second stage of an IV regression with the same specification as reported in Tables VI and VII. For brevity, other coefficients are omitted from the table above, but are included in the underlying regressions. These regressions include only the 361 individuals in the regression sample whose self-reported age and gender match the accounting data, who have no TIAA-CREF IRAs, and for whom either (1) the absolute difference between self-reported TIAA-CREF balances and the TIAA-CREF accounting data is \$10,000 or less, or (2) the absolute log difference between self-reported TIAA-CREF balances and the TIAA-CREF accounting data is .10 log points or less. Asterisks indicate the level of statistical confidence for rejection of the hypothesis that the relevant coefficient is (independently) equal to zero: *** indicates rejection at better than a 1 percent level of confidence; ** indicates rejection at better than a 5 percent level; and * indicates rejection at better than a 10 percent level.

Table VII. Second-stage planning/wealth regression results.

Table VIII. Coefficients using self-reported and TIAA-CREF accounting data.

Read:

- Table VII is the main IV result. Instrumented planning is positive and significant for both wealth measures: 0.324 for net worth and 0.398 for gross financial assets.
- Table VIII restricts attention to households whose self-reported TIAA-CREF balances closely match accounting records. The planning coefficients remain similar under both self-reported and administrative data.
- The IV estimates are larger than the OLS estimates, which is consistent with measurement error in planning or a negative feedback effect from exogenous wealth to the need for planning.

8.5 Tables IX and X: preference controls and budgeting channel

TABLE IX
TWO-STAGE LEAST SQUARES REGRESSION RESULTS INCLUDING PREFERENCE
PARAMETER MEASURES

Variable	Coeff.	S.E.	Pr > t
Second-stage results, net worth			
Planning (IV)	0.393**	0.175	0.026
Discounting	-0.166	0.727	0.819
Precaution	-0.012	0.011	0.273
Risk aversion	0.016	0.077	0.835
Bequest	0.012**	0.006	0.039
Second-stage results, gross financial assets			
Planning (IV)	0.467**	0.188	0.014
Discounting	-0.277	0.779	0.722
Precaution	-0.008	0.011	0.456
Risk aversion	-0.059	0.083	0.474
Bequest	0.011*	0.006	0.068
First-stage results (Dep. variable is Q1a)			
Discounting	-0.235	0.860	0.785
Precaution	0.019	0.012	0.115
Risk aversion	0.165*	0.086	0.056
Bequest	0.017***	0.006	0.005
Vacation planning	0.184***	0.052	0.000
Math confidence	0.185***	0.057	0.001

Source. Authors' calculations based on 2000 and 2001 survey data.
There were 316 observations used in all regressions. For the second-stage regressions (top two panels), the dependent variables are natural logarithms of the given wealth measures (net worth in the top panel, and gross financial assets in the middle panel). See the text for a description of the construction of the preference measures. Other right-hand-side variables included the same controls as in our basic specification: current, past, and future income measures; age and age squared; number of children; and indicators for employment status, occupation, education, marital status, gender and participation in a DB plan (respondent and spouse). For brevity, these coefficients are not reported above. Asterisks indicate the level of statistical confidence for

TABLE X
SECOND-STAGE BUDGETING/WEALTH REGRESSION RESULTS

Variable	Net worth			Gross financial assets		
	Coeff.	S.E.	Pr > t	Coeff.	S.E.	Pr > t
Budgeting variable (IV)	0.209	0.146	0.154	0.277*	0.154	0.072
Log 1999 income	0.348**	0.175	0.048	0.409**	0.185	0.028
Zero 1999 income	1.973**	0.892	0.027	2.375**	0.941	0.012
Log past income	0.257	0.173	0.138	0.210	0.183	0.250
Zero past income	0.902	0.966	0.351	0.111	1.019	0.913
Log future income	0.104	0.110	0.341	0.157	0.116	0.176
Zero future income	0.613	0.495	0.217	0.794	0.523	0.129
Age	0.181***	0.052	0.001	0.144**	0.055	0.010
Age ²	-0.001*	0.001	0.052	-0.001	0.001	0.241
Empl. status						
Working		Omitted			Omitted	
Partially retired	0.221	0.217	0.310	0.349	0.229	0.129
Retired	0.178	0.322	0.580	0.122	0.340	0.720
Occupation						
Faculty		Omitted			Omitted	
Mgmt./sen. admin.	-0.118	0.133	0.377	-0.149	0.141	0.292
Tech./professional	0.108	0.126	0.391	0.109	0.133	0.413
Other	-0.124	0.150	0.411	-0.169	0.159	0.288
Education						
College or below	-0.164	0.122	0.181	-0.293**	0.129	0.024
M.A./professional		Omitted			Omitted	
PhD	-0.127	0.112	0.258	-0.114	0.118	0.335
R. has DB plan	-0.127	0.111	0.250	-0.228*	0.117	0.051
S. has DB plan	-0.154	0.139	0.268	-0.287*	0.146	0.050
Marital status						
Curr. married		Omitted			Omitted	
Prev. married	-0.507***	0.158	0.001	-0.497***	0.166	0.003
Never married	-0.300**	0.140	0.033	-0.185	0.148	0.212
Male respondent	0.054	0.106	0.610	0.154	0.112	0.171
Number of kids	0.056	0.053	0.291	0.035	0.056	0.528
Constant	-3.673***	1.376	0.008	-3.592**	1.452	0.014
N	498			498		

Source. Authors' calculations based on 2000 and 2001 survey data.
Dependent variables are natural logarithms of the quantities listed at head of each set of columns. Asterisks indicate the level of statistical confidence for rejection of the hypothesis that the relevant coefficient is (independently) equal to zero: *** indicates rejection at better than a 1 percent level of confidence; ** indicates rejection at better than a 5 percent level; and * indicates rejection at better than a 10 percent level.

Table IX. 2SLS results including preference-parameter measures.

Table X. Second-stage budgeting/wealth regression results.

Read:

- Table IX shows that planning remains positive and significant after adding measures of discounting, precaution, risk aversion, and bequest motives. This weakens the interpretation that planning is just a proxy for classical preferences.
- Table X replaces planning with budgeting. Instrumented budgeting is positively related to wealth, especially gross financial assets, even though the reduced-form association between budgeting and wealth is negative.
- This supports the paper's self-control interpretation: households with a high propensity to plan are more likely to budget, and budgeting may help restrain spending.

9 Takeaways for future research

The paper has two lasting lessons. First, survey-based behavioral measures can be useful in macro-finance when they are designed around a specific identification problem. Second, models of saving

should take seriously the implementation problem: households may know that they want to save but differ in their ability to make that goal operational.

A useful way to state the paper's contribution is this: **preferences determine desired saving, but planning and budgeting help determine whether the household actually achieves it.**